

Seasonal and regional evaluation of sixteen temperature-based solar radiation models for India using a Global Performance Indicator

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Abstract

Reliable estimation of surface solar radiation is crucial for climatological analyses and land–atmosphere applications; however, direct radiometric observations remain sparse across much of India. Under such data limitations, temperature-based empirical models continue to be widely used, although their seasonal robustness and spatial transferability across diverse climatic regimes remain insufficiently understood. This study assesses the accuracy of temperature-based models in predicting solar radiation required for agricultural planning across India. Daily data on maximum and minimum temperatures and sunshine duration from 14 agrometeorological stations, covering the period 2002–2023, were used to estimate reference solar radiation using the Angström–Prescott model and to validate 16 empirical temperature-input formulations without any local calibration. Model performance was assessed at daily scale using RMSE, MAE, MBE and R^2 , which were synthesized into a Global Performance Indicator (GPI) to obtain season- and location-specific rankings. Results show a clear control of seasonality and geographical setting on model skill: in winter and post-monsoon, the Hargreaves–Samani model (M-1) generally provided the lowest errors across interior and semi-arid stations, whereas humidity-sensitive formulations such as M-5 and the Hassan-based model (M-13) performed better at coastal and high-humidity sites. During the monsoon, when persistent cloud cover weakens the link between diurnal temperature range and radiation, the Rao et al. model (M-2) consistently achieved the highest GPI values at most locations despite overall reduced R^2 . Across all seasons, a small subset of models (notably M-1, M-2 and M-13) emerged as robust and transferable, offering practical, temperature-only options for operational agrometeorological advisories, evapotranspiration estimation and crop simulation in regions lacking direct solar radiation measurements.

1. Introduction

Surface solar radiation is a key driver of the climate system that primary controls land–atmosphere exchanges, evapotranspiration and crop productivity. Solar radiation is the fundamental source of energy driving the Earth's climate system and the primary determinant of agricultural productivity. Accurate estimation of incoming solar radiation at the Earth's surface is essential for numerous applications such as crop growth simulation, evapotranspiration estimation, hydrological modelling, and the improvement of agrometeorological advisories. Furthermore, reliable estimates of surface solar radiation are essential for climate monitoring, hydrological modelling and energy applications, in addition to agricultural uses. Despite its importance, direct measurements of solar radiation using pyranometers remain sparse and unevenly distributed in many parts of the world, particularly across developing regions and complex climatic domains such as the Indian subcontinent. India Meteorological Department (IMD) operates a limited number of Solar Radiation Station Network (46) as compared with more than 547 surface weather observatories and about 223 agrometeorological observatories measuring temperature and humidity (India Meteorological Department 2025), while radiometric stations remain sparse and unevenly distributed. This spatial limitation constrains regional climate analyses and impact assessments that require long-term radiation data in agricultural planning, modelling, and advisory services. Consequently, reliable indirect estimation of solar radiation from routinely measured parameters such as temperature and sunshine duration becomes vital for both operational and research applications. Numerous empirical models have been proposed over the decades to estimate global solar radiation using readily available meteorological variables. Among these, the Angström–Prescott (A–P) model (Ångström, A. 1924; Prescott J. A. 1940), which relates solar radiation to sunshine duration, is one of the earliest and most widely used formulations, as applied in studies like (Kirmani et al. 2015) for New Delhi. However, sunshine duration data are not available at many automatic weather stations, particularly in remote or recently established agrometeorological observatories. Under such circumstances, temperature-based models offer a practical, low-cost alternative, as maximum (T_{max}) and minimum (T_{min}) air

temperatures are recorded at almost all IMD and agricultural research stations across the country. The utility of such empirical approaches is well documented across diverse climatic regions (Supit and Kappel 1998; Ertekin and Yaldız 1999; Samani 2000; Chen et al. 2004; Falayi et al. 2008; Ghazouani et al. 2021; Marif et al. 2022). Comparative analyses, such as the work by (Onyeka et al. 2021) in Sub-Saharan Africa, further highlight the methodological practice of using statistical metrics, such as RMSE and R^2 , to rank the performance of different model types.

The earliest formulations by (Hargreaves and Samani 1982, 1985; Bristow and Campbell 1984) laid the foundation for linking the diurnal temperature range ($\Delta T = T_{\max} - T_{\min}$) with incident solar radiation. Subsequent researchers modified and improved these basic models by incorporating additional parameters, such as mean temperature, extraterrestrial radiation (R_a), relative humidity, elevation, and empirical correction coefficients, to better represent regional climatic conditions. The performance of these models, including foundational ones such as Hargreaves-Samani, has been assessed across various locales, from tropical savannas (Ugwu and Ugwuanyi, A. I. 2011) to specific Indian cities (Panday and Katiyar 2010; Veeraboina et al. 2011), demonstrating their utility but also variable accuracy. Several comparative evaluations conducted in different parts of the world have demonstrated that no single model performs uniformly well across all climatic regions. This is evident in findings from India, where studies such as (Singh et al. 2024) in Jharkhand and (Siva Krishna Rao K et al. 2018) in a tropical Indian climate reported significant model-specific and seasonal variability in accuracy. Therefore, region-specific evaluation and ranking of available models are essential before operational use. Methodologies for such comprehensive ranking have been advanced by Indian researchers, including the use of a synthesized Global Performance Indicator (GPI) by (Jamil et al. 2020; Makade et al. 2020, 2021) to evaluate and rank numerous models across multiple locations. Despite these valuable contributions, a comprehensive and consistent evaluation across India's diverse agro-climatic zones has been lacking. Many past comparisons were confined to a small number of models, specific regions, or used short temporal datasets, which restricts the generalizability of their findings. Large-scale studies such as (Singh et al. 2018) provided crucial methodological frameworks and regional coefficients, while (Makade et al. 2021) reviewed a vast number of models; however, a unified assessment of a curated set of temperature-based models under a consistent, non-calibrated validation protocol across India is needed. Moreover, some models were developed under local calibration conditions, making it difficult to assess their true performance in an independent setting. However, most previous Indian studies have focused on a small number of stations or models, limiting their relevance for national-scale climate applications.

To bridge these gaps, the present study aims to conduct a comprehensive intercomparison of 16 temperature-input-based solar radiation estimation models using long-term (2002–2023) meteorological data from fourteen representative locations across India. The analysis focuses on evaluating the models' ability to reproduce reference solar radiation computed using the Angström–Prescott formulation at daily scales without any local calibration. The large dataset spanning more than two decades and diverse climatic conditions enables the objective identification of the most robust and transferable models for national-level agrometeorological applications. The selection of models in this study was based on a systematic literature review, shortlisting sixteen distinct empirical models based on accessibility, recency, and sole reliance on temperature inputs. This deliberate filtering ensures a fair comparison among models that rely on uniform and widely available meteorological inputs, thereby enhancing the operational relevance of the results. Following methodological precedents set by studies such as (Sivamadhavi and Selvaraj 2012), the evaluation employs multiple statistical indices (RMSE, MAE, MBE, R^2) and an integrated Global Performance Indicator (GPI) to synthesize model performance. Ultimately, this research seeks to move beyond generic recommendations by developing a seasonally and geographically stratified framework for model selection. It aims to identify the most suitable and transferable temperature-based models that can be reliably recommended

for operational agrometeorological advisory services and crop simulation applications across India's varied climates, particularly in regions lacking measured solar radiation data.

2. Materials and Methods

The analysis was conducted using long-term daily meteorological observations from 14 stations distributed across India's major climatic regions, encompassing arid, semi-arid, tropical wet, humid subtropical, coastal, and monsoon-dominated environments. The selected locations, viz., Bhubaneswar, Goa, Guwahati, Hyderabad, Jabalpur, Jaipur, Jammu (Chatha), Jamnagar, Jodhpur, Kottayam, Nagrakata, New Delhi, Pondicherry, and Pune provide a representative coverage of India's pronounced climatic heterogeneity (Fig. 1). Geographical coordinates, elevation, and data availability for each station are summarized in Table 1.

Table 1
Geographical coordinates and data period for the 14 study locations across India

SN	District (Station) State Name	Latitude	Longitude	Altitude	Years_Covered
1	Khordha (Bhubaneswar), Odisha	20.15	85.52	25.9	2002–2023 (22)
2	North Goa (Ela-Goa), Goa	15.3	73.55	60	2002–2023 (22)
3	Kamrup Metropolitan (Guwahati), Assam	26.35	90.52	105	2002–2020 (19)
4	Hyderabad (Hyderabad), Telangana	17.19	78.23	520	2002–2023 (22)
5	Jabalpur (Jabalpur), Madhya Pradesh	23.13	79.57	411	2002–2023 (22)
6	Jaipur (Jaipur), Rajasthan	26.51	75.47	390	2002–2023 (22)
7	Jammu (Chatha), Jammu & Kashmir	32.39	74.54	332	2009–2023 (15)
8	Jamnagar (Jamnagar), Gujarat	22.28	70.04	69	2008–2023 (16)
9	Jodhpur (Jodhpur), Rajasthan	26.18	73.01	224	2002–2023 (22)
10	Kottayam (Kottayam), Kerala	9.32	76.36	77.87	2002–2023 (22)
11	Jalpaiguri (Nagrakata), West Bengal	26.54	88.55	228.6	2002–2023 (22)
12	New Delhi (New Delhi), Delhi	28.04	77.1	220	2003–2023 (21)
13	Karaikal (Pondicherry), Puducherry	10.55	79.49	4	2002–2023 (22)
14	Pune (Pune), Maharashtra	18.32	73.51	559	2002–2023 (22)

2.1 Calculation of extraterrestrial radiation (R_a)

The extraterrestrial radiation (R_a) was calculated for each day of the year and for different latitudes using the solar constant, solar declination, and time of the year. The formula for (R_a) is given as:

$$R_a = \frac{24 \times 60}{\pi} G_{sc} d_r [\omega_s \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \sin(\omega_s)] \tag{1}$$

where R_a is the extra-terrestrial radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$), G_{sc} is the solar constant ($0.0820 \text{ MJ m}^{-2} \text{ day}^{-1}$), d_r is the inverse relative distance between the Earth and the Sun (Eq. 2), δ is the solar declination in radians (Eq. 3), ω_s is the sunset hour angle in radians (Eq. 4), and ϕ is the latitude in radians. The inverse relative distance (d_r), solar declination (δ), and sunset hour angle (ω_s) are calculated as:

$$d_r = 1 + 0.003 \cos \left(\frac{2\pi J}{365} \right) \quad (2)$$

$$\delta = 0.409 \sin \left(\frac{2\pi J}{365} - 1.39 \right) \quad (3)$$

$$\omega_s = \arccos [-\tan(\phi) \tan(\delta)] \quad (4)$$

where J is the day of the year, ranging from 1 (1 January) to 365 or 366 (31 December).

2.2 Benchmark Reference and temperature based Solar Radiation Estimation Models

This study does not claim absolute validation of temperature-based models against true surface irradiance. Instead, it evaluates their ability to reproduce a widely accepted sunshine-based operational standard (A-P), which is currently the most practical alternative in India's agrometeorological network. Hence, Angstrom-Prescott (A-P) model was taken as this reference standard due to its widespread recognition, theoretical robustness, and proven reliability (Singh et al. 2024). The A-P model formula (Eq. 5) estimates solar radiation based on the ratio of actual to maximum possible sunshine hours and extra-terrestrial radiation. The maximum possible sunshine hours (N) were computed using the sunset hour angle (Eq. 6). The A-P model is expressed as:

$$R_s = \left[a + b \left(\frac{n}{N} \right) \right] R_a \quad (5)$$

where n (in some models it is indicated as S) is the actual number of bright sunshine hours (hr), N (in some models it is indicated as S_0) is the maximum possible sunshine hours (hr), R_s (in some models it is indicated as H) is the solar radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$), R_a (in some models it is indicated as H_0) is the extra-terrestrial solar radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$), and a and b are empirical coefficients with values of 0.25 and 0.50, respectively. The maximum possible sunshine hours (N) were computed as:

$$N = \frac{24\omega_s}{\pi} \quad (6)$$

The A-P model was selected for three primary reasons:

1. **Theoretical Robustness:** Its foundation in atmospheric physics, using sunshine duration as a direct proxy for cloud cover, makes it a more physically sound reference than a purely empirical temperature-based model.
2. **Operational Data Context:** Sunshine data is more widely available at Indian agrometeorological stations than solar radiation measurements, making the A-P model a relevant and practical standard for operational advisory services.
3. **Comparative Goal:** The core objective is a relative performance assessment to identify which temperature-based models can best replicate the radiation estimates of this widely accepted sunshine-based model. This directly tests the operational utility of temperature data as a substitute where sunshine records are unavailable.

Therefore, this study evaluates the potential of temperature-based models to serve as reliable proxies for the established A–P standard, with significant practical value for regions with limited meteorological data. The performance of the 16 temperature-based models, detailed in Table 2, was analysed by comparing their outputs to the benchmark A-P model estimates.

Table 2
Empirical temperature-based models for solar radiation estimation evaluated in the study.

SN	Model Name	Model Formula	Model Code
1	Hargreaves & Samani (1985), used as-is	$R_s = 0.16 \sqrt{(T_{\max} - T_{\min})} R_a$	M-1
2	Developed by Rao et al. (2017)	$H = 1.29 + 0.3626 * (T_{\max} - T_{\min})$	M-2
3	Richardson (2018), recalibrated by Marif et al. (2022)	$H = 0.3124(\Delta T)^{0.2942}$	M-3
4	Sarkar & Sifat (2016), recalibrated by Marif et al. (2022)	$H = 0.7730 - 0.0296 (\Delta T) + 0.0016(\Delta T)^2$	M-4
5	Chen et al. (2004), recalibrated by Marif et al. (2022)	$H = 0.2063 + 0.1801 \ln (\Delta T)$	M-5
6	Hargreaves & Riley (1985), recalibrated by Marif et al. (2022)	$H = 0.3040 + 0.1008 \sqrt{\Delta T}$	M-6
7	Developed by Panday & Katiyar (2010)	$H = 0.2889 + 0.1562 \left(\frac{T_{\max}}{T_{\min}} \right)$	M-7
8	Developed by Falayi et al. (2008)	$H = 1.7217 - 1.691 (\Theta)$	M-8
9	Chen et al. (2004), used as-is	$R_s = [0.28 \ln (T_{\max} - T_{\min}) + 0.15] R_a$	M-9
10	Ertekin & Yaldiz (1999), used as-is	$R_s = -4.46 + 0.477 R_a - 0.226 T; (T = \text{Meantemperature})$	M-10
11	Developed by Sivamadhavi & Selvaraj (2012)	$H = 0.4610 + 0.0029 \Delta T$	M-11
12	Developed by Sivamadhavi & Selvaraj (2012)	$H = 0.5290 + 0.0516 \times \theta \left(\text{where } \theta = \frac{T_{\max}}{T_{\min}} \right)$	M-12
13	Hassan et al. (2016), recalibrated by Ghazouani et al. (2022)	$H = 0.206590 + 0.026851 T_{\max} - 0.024184 T_{\min}$	M-13
14	Developed by Kirmani et al. (2015)	$H = 0.85235 - 0.00997 (T); (T = \text{Meantemperature})$	M-14
15	Hargreaves & Samani (1982), recalibrated by Onyeka et al. (2021)	$H = 0.236 + 0.106 \Delta T^{0.5}$	M-15

SN	Model Name	Model Formula	Model Code
16	Samani (2000) used as-is	$\hat{H} = 0.00185\Delta T^2 - 0.0433\Delta T + 0.4023$	M-16

2.3 Statistical Analysis:

The accuracy and performance of all models were evaluated using a comprehensive set of statistical metrics, detailed in Table 3. These included the Mean Absolute Error (MAE) to measure the average magnitude of errors, the Mean Bias Error (MBE) to assess systematic bias, and the Root Mean Squared Error (RMSE) to quantify the dispersion of prediction errors. The Coefficient of Determination (R²) was used to evaluate the proportion of variance in the observations explained by the models. Depending on the statistical error test, each test gives a different prediction of the best model. To overcome this limitation and synthesize these multiple performance measures into a single metric, a Global Performance Indicator (GPI) was calculated. The Statistical package R was used for the analysis.

Table 3
Statistical performance metrics used for model validation and their ideal values.

S.N.	Statistical Test Name	Equation	Ideal Value
1	Mean Absolute Error (MAE)	$MAE = \left(\frac{1}{n}\right) \times \sum H_{exp} - H_{pred} $	0 (lower is better)
2	Root Mean Square Error (RMSE)	$RMSE = \sqrt{\left[\left(\frac{1}{n}\right) \times \sum (H_{exp} - H_{pred})^2\right]}$	0 (lower is better)
3	Mean Bias Error (MBE)	$MBE = \left(\frac{1}{n}\right) \times \sum (H_{exp} - H_{pred})$	0 (closer to 0 better)
4	Coefficient of Determination (R ²)	$R^2 = 1 - \left[\frac{\sum (H_{exp} - H_{pred})^2}{\sum (H_{exp} - \bar{H})^2}\right]$	1 (higher is better)
5	Global Performance Indicator (GPI)	$GPI_i = \sum_{j=1}^n \alpha_j \left(\tilde{y}_j - y_{ij}\right)$	Higher is better

2.3.1 Global Performance Indicator (GPI) as a Ranking Tool

Given the multi-criteria nature of model evaluation, where a model can perform well in one statistical metric (e.g., R²) and poorly in another (e.g., MBE), a robust method for synthesizing these measures into a single, unambiguous ranking was essential. For this purpose, the Global Performance Indicator (GPI) was adopted and calculated for each model (Despotovic et al. 2015).

The GPI was established using a systematic ranking procedure. Initially, the values of all statistical indicators (R², RMSE, MAE, MBE) for all models were normalized to a range of 0 to 1. This scaling ensures all metrics are

dimensionless and comparable. The core principle of the GPI is to measure the deviation of each model's performance from the median performance across all models for each statistical indicator.

Mathematically, for the i th model, the GPI is calculated as:

$$GPI_i = \sum_{j=1}^n \alpha_j (\tilde{y}_j - y_{ij})$$

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Where $\alpha_j = \begin{cases} -1, & \text{for } R^2 \\ +1 & \text{for all other errors (RMSE, MAE, MBE)} \end{cases}$ and,

$\tilde{y}_j$ is the median of scaled values and

y_{ij} is the scaled value of j th indicator for i th model

This formulation means that for a model to achieve a high GPI score, it must simultaneously: Have an R^2 value higher than the median (the negative weight inverts the subtraction, making a higher value beneficial). Have error values (RMSE, MAE, MBE) lower than the median. The GPI, therefore, provides a holistic and balanced score. A higher positive GPI value indicates a model's performance is superior to the median model across the entire suite of metrics, making it the most robust and reliable choice. This method effectively avoids the pitfall of selecting a model that excels in one aspect but fails in others.

3. Results and discussion

The comprehensive evaluation of sixteen temperature-based solar radiation (SR) estimation models across fourteen climatically diverse Indian locations over a 22-year period (2002–2023) yielded a detailed and seasonally differentiated performance matrix. The primary objective of this analysis was to move beyond generic or single-model recommendations and to develop a nuanced, evidence-based model selection framework tailored to India's heterogeneous agro-climatic environments and strong seasonal contrasts. The comparative assessment, synthesized using multiple statistical indicators (RMSE, MAE, MBE, R^2) and the integrated Global Performance Indicator (GPI), unequivocally demonstrates that model efficacy is not universal. Instead, it is intrinsically governed by seasonal atmospheric dynamics, temperature variability, humidity regime, cloud characteristics, and geographical context. The following sections present a season-wise synthesis of these findings, supported by quantitative evidence from Table 4, and provide definitive guidance for operational deployment.

3.1 Seasonal Dynamics in Model Performance

A central outcome of this study is the pronounced influence of seasonality on the performance of temperature-based SR estimation models. The optimal model varies markedly between seasons, underscoring the necessity of a dynamic and stratified modelling approach for operational agrometeorology. The discussion below synthesizes the performance of the top-ranked models (based on GPI) across all fourteen locations for each season.

3.1.1 Winter Season (January–February)

Winter represents the most atmospherically stable period across much of India, characterized by relatively clear skies, low humidity, and enhanced diurnal temperature range. Under these conditions, temperature-based models

achieved their highest accuracy, reflected by elevated R^2 values and reduced error magnitudes. Across most interior and continental stations, the Hargreaves and Samani model (M-1) consistently emerged as the best-performing formulation. For instance, at Jabalpur, M-1 achieved an R^2 of 0.66, with a low RMSE of $2.42 \text{ MJ m}^{-2} \text{ day}^{-1}$ and MAE of $1.57 \text{ MJ m}^{-2} \text{ day}^{-1}$, indicating excellent agreement with the Angström–Prescott (A–P) reference radiation. Similarly, New Delhi and Guwahati recorded R^2 values of 0.51 and 0.34, respectively, with RMSE values remaining within acceptable operational limits. The relatively stable, moderately negative MBE values observed across these stations suggest a consistent but mild underestimation, typical when clear-sky radiation slightly exceeds temperature-derived approximations. In contrast, stations influenced by maritime or humid air masses exhibited a shift in optimal model choice. At Goa, alternative formulations such as M-5 and M-13 demonstrated superior winter performance, with RMSE values close to $2 \text{ MJ m}^{-2} \text{ day}^{-1}$, reflecting their improved sensitivity to humidity-modulated transmissivity. In plateau and semi-arid environments such as Pune, refined non-linear temperature formulations (e.g., M-11 and M-13) outperformed simpler models, with RMSE values falling below $2.0 \text{ MJ m}^{-2} \text{ day}^{-1}$ and R^2 exceeding 0.65. These results confirm that under clear-sky winter conditions, models that incorporate non-linear temperature–radiation relationships can better capture subtle variations in atmospheric transmissivity. Overall, winter emerged as the most favourable season for temperature-based SR estimation. The dominance of M-1 in inland regions, alongside the strong performance of modified formulations in coastal and plateau regions, suggests that winter conditions amplify the physical linkage between the diurnal temperature range and incoming solar radiation.

3.1.2 Pre-Monsoon Season (March–May)

The pre-monsoon season introduces rising temperatures, enhanced aerosol loading, dust transport, and episodic humidity increases, all of which weaken the direct relationship between temperature variability and surface solar radiation. Consequently, greater dispersion in model performance was observed, with no single formulation dominating all locations. Across several central and semi-arid stations, including Jabalpur and Hyderabad, modified and non-linear temperature models such as M-11 and M-13 consistently ranked among the top performers. For example, RMSE values at these stations typically ranged between 3.0 and $4.0 \text{ MJ m}^{-2} \text{ day}^{-1}$, reflecting the models' improved ability to accommodate non-linear attenuation effects associated with dust and aerosol accumulation. Coastal locations exhibited a distinct response pattern. At Goa and Pondicherry, humidity-sensitive formulations (notably M-5) outperformed simpler ΔT -based models, achieving RMSE values of approximately 3.5 – $4.5 \text{ MJ m}^{-2} \text{ day}^{-1}$. These results indicate that as pre-monsoon humidity and convective cloud development increase, models incorporating mean-temperature or ratio-based terms provide more stable estimates. In humid and forest-influenced eastern stations such as Guwahati, classical models including M-1 remained competitive but with noticeably higher errors (RMSE often exceeding $4.5 \text{ MJ m}^{-2} \text{ day}^{-1}$), reflecting the progressive weakening of the ΔT –SR relationship. Overall, the pre-monsoon results emphasize that atmospheric heterogeneity during this season diminishes the universal applicability of classical temperature-based models, necessitating the use of formulations capable of representing non-linear radiative attenuation.

3.1.3 Monsoon Season (June–September)

The monsoon season represents the most challenging atmospheric environment for temperature-based SR estimation due to persistent cloud cover, high humidity, and frequent rainfall. Under these conditions, the physical linkage between temperature variability and solar radiation is substantially weakened, resulting in generally low R^2 values (typically 0.07–0.34) across all stations. Despite these limitations, the Rao et al. formulation (M-2) consistently achieved the highest GPI scores during the monsoon across the majority of locations. For example, at New Delhi, M-2 produced an RMSE of $4.87 \text{ MJ m}^{-2} \text{ day}^{-1}$ with near-zero MBE, while at Guwahati and Kottayam,

RMSE values of approximately 4.4 and 3.7 MJ m⁻² day⁻¹, respectively, highlighted its relative robustness under cloud-dominated conditions. These results suggest that M-2 is better structured to withstand the decoupling between surface temperature variation and incoming radiation that characterizes Indian monsoon dynamics. The Hargreaves and Samani model (M-1) maintained reasonable stability at several stations, including Pune, Hyderabad, and Bhubaneshwar, with RMSE values generally between 3.5 and 4.5 MJ m⁻² day⁻¹. Although its predictive power was reduced relative to dry seasons, its continued competitiveness underscores its inherent robustness even when ΔT provides limited radiative information. Overall, the monsoon analysis clearly demonstrates that while temperature-based models face inherent limitations during this season, carefully selected formulations such as M-2 can still provide the most reliable estimates available under data-scarce operational conditions.

3.1.4 Post-Monsoon Season (October–December)

The post-monsoon season marks a rapid transition from humid, cloud-dominated conditions to clearer and more stable atmospheric states. As cloudiness diminishes and humidity declines, the temperature–radiation relationship strengthens, leading to a marked improvement in model performance relative to the monsoon period. In several coastal and eastern locations, including Goa, Bhubaneshwar, and Guwahati, the Hassan-modified formulation (M-13) emerged as the top-ranked model. At Bhubaneshwar, for instance, M-13 achieved an RMSE of 2.92 MJ m⁻² day⁻¹ and MAE of 2.29 MJ m⁻² day⁻¹, reflecting its strong adaptability to residual moisture and intermittent cloud conditions typical of post-monsoon environments. Conversely, in interior and semi-arid regions such as Pune, Jammu, and Jodhpur, the Hargreaves and Samani model (M-1) regained dominance. At Pune, RMSE values dropped to approximately 2.2 MJ m⁻² day⁻¹, while Jammu recorded R² values approaching 0.67, indicating highly accurate representation of post-monsoon solar transmissivity under clear-sky conditions. Overall, the post-monsoon results reveal a clear dual pattern: humidity-sensitive formulations perform best in coastal and eastern regions, whereas ΔT -driven models dominate in inland and arid zones, reflecting strong radiative clarity following monsoon withdrawal.

3.2 Geographical Influence on Model Efficacy

Beyond seasonality, geographical setting exerted a distinct influence on model suitability. Coastal and high-humidity zones (e.g., Goa, Bhubaneshwar, Kottayam) consistently favoured humidity-responsive formulations such as M-5 and M-13, while arid and semi-arid regions (e.g., Jodhpur and Jaipur) benefited from clear-sky dominance, enabling ΔT -based models like M-1 to achieve high precision. Interior continental locations (e.g., Hyderabad, Jabalpur, and New Delhi) further validated the broad transferability and robustness of M-1, establishing it as a reliable default option for large agricultural regions.

3.3 Validation of Practical Transferability

A key contribution of this study lies in demonstrating that reliable SR estimates can be achieved without local calibration. The consistent high-ranking performance of models such as M-1, M-2, and M-13 across diverse climatic regimes confirms their inherent transferability. This finding has immediate practical relevance for agrometeorological services, as it eliminates the need for data-intensive calibration procedures. By adopting season- and region-specific model recommendations derived from the GPI framework, practitioners can substantially improve solar radiation inputs for evapotranspiration estimation, irrigation scheduling, crop simulation, and climate–energy planning. While statistical indicators provide quantitative measures of model skill, graphical comparison further illustrates temporal agreement. Accordingly, Fig. 2 presents the mean-monthly variation of estimated solar radiation from the top three ranked models alongside the Angström–Prescott reference for each study location, visually confirming the seasonal fidelity of the recommended formulations.

Table 4
Ranking of the top three best-performing models based on the Global Performance Indicator (GPI)
across different locations and seasons.

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
1. Bhubaneswar	Winter	Rank-1	M-1	0.48	2.81	2.09	-0.92	0.66
		Rank-2	M-13	0.47	3.28	2.50	-1.87	0.48
		Rank-3	M-7	0.39	3.01	2.24	-0.56	0.42
	Pre-Monsoon	Rank-1	M-11	0.11	3.33	2.59	0.53	0.50
		Rank-2	M-5	0.13	4.68	3.58	-3.32	0.32
		Rank-3	M-12	0.08	3.35	2.48	0.06	0.27
	Monsoon	Rank-1	M-1	0.27	4.45	3.69	-0.56	0.99
		Rank-2	M-2	0.27	4.70	3.66	1.50	0.90
		Rank-3	M-8	0.23	4.80	3.82	0.52	0.80
	Post-Monsoon	Rank-1	M-13	0.45	2.92	2.29	-0.53	0.83
		Rank-2	M-1	0.43	2.91	2.31	0.34	0.81
		Rank-3	M-3	0.49	3.81	2.85	-2.44	0.64
	Annual	Rank-1	M-1	0.43	3.63	2.86	-0.18	0.97
		Rank-2	M-13	0.41	4.01	3.13	-1.55	0.79
		Rank-3	M-5	0.40	5.28	4.08	-3.75	0.43
2. Goa	Winter	Rank-1	M-5	0.51	1.87	1.26	-1.02	0.60
		Rank-2	M-3	0.50	1.87	1.28	-1.00	0.57
		Rank-3	M-6	0.50	1.90	1.30	-1.06	0.56
	Pre-Monsoon	Rank-1	M-5	0.13	4.09	2.97	-2.17	0.46
		Rank-2	M-3	0.12	4.02	2.92	-2.00	0.39
		Rank-3	M-6	0.11	4.17	3.01	-2.27	0.32
	Monsoon	Rank-1	M-2	0.16	4.33	3.37	0.30	1.42
		Rank-2	M-13	0.15	5.29	4.60	-3.22	0.91
		Rank-3	M-8	0.09	5.27	4.16	-1.57	0.87
	Post-Monsoon	Rank-1	M-13	0.29	2.93	2.22	-0.14	0.53
		Rank-2	M-1	0.29	3.66	2.74	-2.11	0.30
		Rank-3	M-5	0.24	3.59	2.41	-1.95	0.18
	Annual	Rank-1	M-13	0.40	3.91	3.03	-0.97	0.76
		Rank-2	M-1	0.41	4.72	3.66	-2.88	0.55

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
3. Guwahati	Winter	Rank-3	M-8	0.33	4.61	3.68	-1.20	0.45
		Rank-1	M-1	0.34	3.67	2.73	-2.33	1.00
		Rank-2	M-13	0.33	4.05	3.09	-2.81	0.85
	Pre-Monsoon	Rank-3	M-11	0.19	3.12	2.44	-0.23	0.63
		Rank-1	M-1	0.23	4.72	3.70	-2.78	1.19
		Rank-2	M-13	0.23	5.61	4.51	-4.03	0.95
	Monsoon	Rank-3	M-7	0.10	4.86	3.79	-2.76	0.65
		Rank-1	M-2	0.17	4.38	3.48	-1.18	1.35
		Rank-2	M-1	0.19	4.88	4.01	-2.84	1.22
	Post-Monsoon	Rank-3	M-13	0.20	5.79	4.85	-4.26	0.96
		Rank-1	M-1	0.40	2.53	1.87	-0.80	1.04
		Rank-2	M-13	0.41	2.77	2.08	-1.41	0.97
	Annual	Rank-3	M-3	0.41	3.93	3.18	-3.11	0.58
		Rank-1	M-1	0.38	4.14	3.16	-2.20	1.25
		Rank-2	M-13	0.38	4.83	3.75	-3.21	1.04
	Annual	Rank-3	M-7	0.25	4.77	3.74	-2.89	0.68
4. Hyderabad	Winter	Rank-1	M-1	0.54	1.97	1.43	-0.61	0.77
		Rank-2	M-5	0.59	2.92	2.40	-2.35	0.71
		Rank-3	M-3	0.59	2.98	2.49	-2.43	0.68
	Pre-Monsoon	Rank-1	M-1	0.18	2.98	2.27	-1.12	0.72
		Rank-2	M-5	0.21	4.43	3.66	-3.61	0.56
		Rank-3	M-3	0.21	4.47	3.71	-3.65	0.53
	Monsoon	Rank-1	M-2	0.32	4.32	3.46	0.80	0.83
		Rank-2	M-1	0.31	4.32	3.54	-0.87	0.80
		Rank-3	M-13	0.31	4.79	3.92	-2.20	0.60
	Post-Monsoon	Rank-1	M-1	0.43	2.58	1.96	-0.35	0.79
		Rank-2	M-13	0.41	2.96	2.30	-1.33	0.61
		Rank-3	M-3	0.44	3.81	2.94	-2.80	0.48
	Annual	Rank-1	M-1	0.51	3.28	2.48	-0.76	0.87
		Rank-2	M-13	0.49	4.01	3.17	-2.20	0.65
		Rank-3	M-3	0.45	5.07	3.96	-3.79	0.34

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI	
5. Jabalpur	Winter	Rank-3	M-7	0.27	3.91	3.03	-0.54	0.34	
		Rank-1	M-1	0.66	2.42	1.57	-1.02	0.80	
		Rank-2	M-13	0.64	2.84	2.09	-1.77	0.64	
		Rank-3	M-3	0.62	3.63	2.79	-2.77	0.42	
		Pre-Monsoon	Rank-1	M-1	0.28	4.04	3.22	-2.76	0.70
			Rank-2	M-5	0.29	5.49	4.73	-4.72	0.46
	Rank-3		M-3	0.29	5.61	4.88	-4.86	0.45	
	Monsoon	Rank-1	M-2	0.44	3.88	3.01	0.71	1.17	
		Rank-2	M-1	0.42	3.98	3.24	-1.10	1.05	
		Rank-3	M-8	0.38	4.37	3.42	-0.92	0.93	
	Post-Monsoon	Rank-1	M-1	0.45	2.56	1.89	-0.88	0.55	
		Rank-2	M-5	0.60	3.47	2.77	-2.70	0.53	
		Rank-3	M-3	0.59	3.51	2.82	-2.74	0.51	
	Annual	Rank-1	M-1	0.58	3.44	2.60	-1.44	1.11	
		Rank-2	M-13	0.56	4.49	3.56	-2.92	0.82	
		Rank-3	M-5	0.52	5.30	4.25	-4.16	0.56	
6. Jaipur	Winter	Rank-1	M-5	0.56	2.64	1.72	-1.42	0.62	
		Rank-2	M-1	0.53	2.37	1.85	0.70	0.62	
		Rank-3	M-3	0.56	2.62	1.73	-1.38	0.62	
	Pre-Monsoon	Rank-1	M-5	0.32	3.56	2.45	-2.18	0.28	
		Rank-2	M-3	0.31	3.55	2.44	-2.13	0.26	
		Rank-3	M-6	0.31	3.60	2.48	-2.22	0.25	
	Monsoon	Rank-1	M-13	0.34	4.50	3.64	-0.80	0.55	
		Rank-2	M-1	0.34	4.49	3.77	0.87	0.54	
		Rank-3	M-2	0.32	5.16	4.24	2.50	0.32	
	Post-Monsoon	Rank-1	M-5	0.57	2.51	1.60	-1.36	0.33	
		Rank-2	M-13	0.51	2.29	1.60	-0.15	0.32	
		Rank-3	M-3	0.56	2.51	1.62	-1.34	0.32	
	Annual	Rank-1	M-13	0.57	3.44	2.52	-0.50	0.65	
		Rank-2	M-1	0.57	3.39	2.59	0.75	0.64	
		Rank-3	M-5	0.55	4.11	2.73	-2.32	0.46	

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
7. Jammu (Chatha)	Winter	Rank-1	M-1	0.50	3.57	2.76	-2.34	0.91
		Rank-2	M-13	0.50	3.65	2.86	-2.45	0.90
		Rank-3	M-3	0.47	5.10	4.35	-4.30	0.45
	Pre-Monsoon	Rank-1	M-11	0.29	4.42	3.58	-0.56	0.76
		Rank-2	M-1	0.44	6.06	4.80	-4.58	0.69
		Rank-3	M-12	0.25	4.70	3.59	-1.34	0.58
	Monsoon	Rank-1	M-1	0.19	5.04	3.93	-1.47	0.62
		Rank-2	M-2	0.16	5.27	4.13	-0.43	0.49
		Rank-3	M-13	0.19	5.98	4.61	-3.17	0.36
	Post-Monsoon	Rank-1	M-1	0.67	2.73	2.01	-1.63	0.59
		Rank-2	M-13	0.68	3.05	2.40	-2.19	0.49
		Rank-3	M-12	0.44	2.79	2.21	-0.15	0.34
	Annual	Rank-1	M-11	0.50	4.24	3.31	-1.03	0.68
		Rank-2	M-1	0.57	4.68	3.50	-2.46	0.63
		Rank-3	M-12	0.48	4.57	3.41	-1.81	0.53
8. Jamnagar	Winter	Rank-1	M-5	0.17	3.55	1.81	-1.32	0.37
		Rank-2	M-3	0.17	3.55	1.84	-1.27	0.34
		Rank-3	M-6	0.17	3.57	1.84	-1.34	0.34
	Pre-Monsoon	Rank-1	M-12	0.32	4.12	3.80	3.01	0.69
		Rank-2	M-11	0.33	4.58	4.34	3.64	0.65
		Rank-3	M-4	0.16	3.20	1.78	-0.83	0.45
	Monsoon	Rank-1	M-1	0.35	4.60	3.88	0.97	0.71
		Rank-2	M-13	0.32	4.65	4.01	-0.76	0.64
		Rank-3	M-2	0.38	5.43	4.42	3.18	0.52
	Post-Monsoon	Rank-1	M-5	0.35	2.57	1.43	-0.76	0.58
		Rank-2	M-3	0.34	2.57	1.45	-0.69	0.56
		Rank-3	M-6	0.34	2.61	1.44	-0.82	0.54
	Annual	Rank-1	M-13	0.40	3.95	3.04	0.42	0.85
		Rank-2	M-1	0.43	4.20	3.36	1.70	0.80
		Rank-3	M-5	0.41	4.32	2.79	-1.85	0.77
9. Jodhpur	Winter	Rank-1	M-5	0.49	2.33	1.51	-1.14	0.50

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
	Pre-Monsoon	Rank-2	M-3	0.48	2.35	1.56	-1.13	0.47
		Rank-3	M-6	0.48	2.38	1.57	-1.20	0.45
		Rank-1	M-5	0.35	3.17	2.25	-2.04	0.43
		Rank-2	M-6	0.34	3.23	2.31	-2.11	0.39
		Rank-3	M-3	0.34	3.18	2.28	-2.02	0.39
	Monsoon	Rank-1	M-13	0.35	3.97	3.27	0.03	0.49
		Rank-2	M-1	0.36	4.30	3.67	1.79	0.39
		Rank-3	M-5	0.34	4.78	3.51	-2.63	0.25
	Post-Monsoon	Rank-1	M-5	0.72	1.91	1.33	-1.08	0.45
		Rank-2	M-3	0.71	1.94	1.38	-1.09	0.43
		Rank-3	M-6	0.70	1.99	1.40	-1.16	0.41
	Annual	Rank-1	M-13	0.58	3.17	2.45	-0.38	0.50
		Rank-2	M-1	0.59	3.18	2.47	1.08	0.47
		Rank-3	M-5	0.61	3.47	2.33	-1.86	0.45
10. Kottayam	Winter	Rank-1	M-13	0.28	2.79	2.08	-0.02	0.36
		Rank-2	M-1	0.29	3.02	2.56	1.16	0.26
		Rank-3	M-3	0.29	3.44	2.25	-1.99	0.23
	Pre-Monsoon	Rank-1	M-1	0.24	3.53	2.85	0.16	0.37
		Rank-2	M-13	0.24	3.73	2.92	-1.22	0.29
		Rank-3	M-2	0.22	3.85	3.12	1.18	0.21
	Monsoon	Rank-1	M-2	0.34	3.70	3.02	0.00	1.35
		Rank-2	M-1	0.34	4.09	3.54	-1.63	1.11
		Rank-3	M-13	0.36	4.75	4.14	-2.89	0.95
	Post-Monsoon	Rank-1	M-1	0.33	3.48	2.88	-0.69	0.83
		Rank-2	M-13	0.35	3.83	3.08	-1.75	0.74
		Rank-3	M-2	0.31	3.77	3.03	-1.61	0.66
	Annual	Rank-1	M-1	0.43	3.63	3.03	-0.47	0.80
		Rank-2	M-2	0.38	3.68	2.95	-0.22	0.71
		Rank-3	M-13	0.43	3.99	3.22	-1.70	0.67
11. Nagrakata	Winter	Rank-1	M-1	0.16	4.13	3.09	-2.54	0.84
		Rank-2	M-13	0.17	4.45	3.38	-2.96	0.78

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
	Pre-Monsoon	Rank-3	M-11	0.05	3.40	2.76	-0.46	0.47
		Rank-1	M-1	0.08	5.46	4.21	-2.85	0.89
		Rank-2	M-11	0.06	4.88	3.85	-2.06	0.84
	Monsoon	Rank-3	M-13	0.08	6.18	4.81	-3.94	0.73
		Rank-1	M-1	0.09	5.61	4.44	-2.85	1.19
		Rank-2	M-2	0.07	5.65	4.25	-1.30	1.18
	Post-Monsoon	Rank-3	M-13	0.09	6.43	5.21	-4.28	0.88
		Rank-1	M-1	0.26	3.13	2.38	-0.48	0.56
		Rank-2	M-13	0.27	3.31	2.47	-1.11	0.49
	Annual	Rank-3	M-5	0.32	4.00	2.97	-2.70	0.42
		Rank-1	M-1	0.21	4.81	3.63	-2.19	1.14
		Rank-2	M-13	0.22	5.41	4.11	-3.17	0.97
		Rank-3	M-11	0.13	5.19	4.13	-2.20	0.67
		Rank-1	M-1	0.51	3.90	3.13	-2.72	0.81
		Rank-2	M-13	0.50	4.11	3.34	-2.96	0.74
12. New Delhi	Winter	Rank-3	M-11	0.36	3.39	2.88	-0.63	0.66
		Rank-1	M-1	0.26	4.69	3.64	-3.12	0.63
		Rank-2	M-11	0.17	3.57	2.88	0.49	0.55
	Pre-Monsoon	Rank-3	M-12	0.15	3.55	2.71	-0.09	0.51
		Rank-1	M-2	0.17	4.87	3.84	0.00	0.96
		Rank-2	M-1	0.16	4.95	4.00	-1.49	0.77
	Monsoon	Rank-3	M-8	0.13	5.50	4.32	-1.27	0.54
		Rank-1	M-11	0.42	3.32	2.69	-0.94	0.90
		Rank-2	M-12	0.40	3.66	2.84	-1.70	0.75
	Post-Monsoon	Rank-3	M-1	0.41	4.70	3.79	-3.49	0.45
		Rank-1	M-1	0.53	4.66	3.71	-2.62	0.72
		Rank-2	M-11	0.43	4.50	3.49	-1.43	0.67
	Annual	Rank-3	M-12	0.41	4.84	3.63	-2.16	0.52
		Rank-1	M-5	0.43	3.05	2.32	0.34	0.47
		Rank-2	M-3	0.42	3.10	2.42	0.48	0.44
13. Pondicherry	Winter	Rank-3	M-6	0.41	3.14	2.31	0.09	0.44

Location	Time Period	Rank	Model	R2	RMSE	MAE	MBE	GPI
	Pre-Monsoon	Rank-1	M-5	0.13	3.59	2.50	-0.44	0.47
		Rank-2	M-3	0.11	3.59	2.56	-0.26	0.39
		Rank-3	M-6	0.11	3.65	2.50	-0.61	0.31
	Monsoon	Rank-1	M-13	0.24	4.55	3.68	-0.67	0.32
		Rank-2	M-2	0.23	4.87	4.18	1.94	0.14
		Rank-3	M-1	0.23	5.20	3.91	-2.65	0.11
	Post-Monsoon	Rank-1	M-1	0.37	3.95	3.26	-0.06	0.44
		Rank-2	M-5	0.36	4.36	3.45	-1.71	0.26
		Rank-3	M-2	0.40	4.58	3.84	2.41	0.25
	Annual	Rank-1	M-1	0.32	4.26	3.29	-0.56	0.29
		Rank-2	M-5	0.31	4.53	3.29	-1.63	0.17
		Rank-3	M-3	0.30	4.52	3.32	-1.51	0.14
14. Pune	Winter	Rank-1	M-1	0.66	1.94	1.56	-1.30	0.70
		Rank-2	M-5	0.68	2.88	2.54	-2.53	0.57
		Rank-3	M-3	0.68	3.03	2.72	-2.71	0.55
	Pre-Monsoon	Rank-1	M-5	0.20	4.10	3.35	-3.16	0.53
		Rank-2	M-6	0.18	4.26	3.52	-3.33	0.39
		Rank-3	M-3	0.17	4.22	3.49	-3.27	0.37
	Monsoon	Rank-1	M-1	0.53	3.58	2.99	-0.32	0.97
		Rank-2	M-8	0.50	3.73	2.98	-0.95	0.86
		Rank-3	M-2	0.51	3.91	3.00	1.64	0.81
	Post-Monsoon	Rank-1	M-1	0.53	2.16	1.60	-0.54	0.84
		Rank-2	M-13	0.51	2.70	2.18	-1.61	0.64
		Rank-3	M-3	0.52	3.50	2.80	-2.73	0.49
	Annual	Rank-1	M-1	0.64	2.94	2.30	-0.71	0.89
		Rank-2	M-13	0.61	3.80	3.17	-2.25	0.63
		Rank-3	M-5	0.61	4.66	3.73	-3.57	0.46

4. Conclusion

The present study comprehensively evaluated sixteen temperature-based solar radiation (SR) estimation models across fourteen climatically diverse locations in India using a multi-seasonal dataset spanning 2002–2023. The comparative assessment, carried out using RMSE, MAE, MBE, R², and the integrated Global Performance Indicator

(GPI), clearly demonstrates that no single temperature-based model performs uniformly across all seasons and regions. Instead, model effectiveness is strongly conditioned by prevailing atmospheric conditions, seasonal cloudiness, humidity regime, temperature variability, and geographical characteristics. By explicitly examining season-wise performance patterns, this study establishes a scientifically sound and operationally relevant framework for selecting suitable empirical SR estimation models for agrometeorological services and climate applications, and assessments under data-scarce conditions.

Across all stations, the Hargreaves and Samani model (M-1) consistently demonstrated robust and stable performance, particularly during winter, post-monsoon, and annual evaluations, confirming its status as the most geographically transferable and operationally simple model for inland and semi-arid regions of India. Its superior performance during these seasons arises from relatively clear-sky atmospheric conditions and a pronounced diurnal temperature range, which strengthens the physical linkage between ΔT and surface solar radiation. In cities such as Jabalpur, Pune, Jammu, and New Delhi, M-1 achieved strong statistical accuracy, with RMSE values generally ranging between 2.0 and 3.0 MJ m⁻² day⁻¹, accompanied by stable MAE and low systematic bias. The consistently high annual GPI rankings further validate M-1 as a dependable baseline model for locations experiencing moderate to high sky clearness over a substantial part of the year, even in the absence of local calibration. However, the results also highlight that India's climatic heterogeneity ranging from humid coastal belts to inland plains and semi-arid interiors—necessitates region-specific model selection rather than reliance on a single universal formulation. Coastal and humid locations, particularly Goa, Pondicherry, and Bhubaneswar, exhibited better performance from alternative temperature-based formulations, such as M-5 and M-13, which incorporate linear, ratio-based, or modified temperature relationships. These models demonstrated relatively lower RMSE (approximately 2.8–4.0 MJ m⁻² day⁻¹ across seasons) and balanced MBE, indicating enhanced sensitivity to humidity-driven attenuation and intermittent cloudiness typical of maritime environments. This behaviour underscores that coastal climates require models capable of partially accounting for moisture-induced transmissivity reduction, where temperature range alone may not sufficiently represent radiative variability. In plateau, central, and semi-arid regions such as Pune, Jabalpur, and Hyderabad, several non-linear and modified temperature formulations (including M-11 and M-13) performed competitively during the winter and pre-monsoon seasons. These models captured radiation variability more effectively under conditions influenced by aerosol loading, dust transport, and strong temperature gradients. Their improved performance during dry and thermally dynamic periods suggests that refined temperature–radiation transformations offer added value where atmospheric stability is high but temperature amplitude is modulated by elevation and land–atmosphere interactions.

The monsoon season emerged as the most challenging period for temperature-based SR estimation due to persistent cloud cover, high humidity, and frequent rainfall, which significantly weaken the correlation between temperature variability and solar radiation. Under these conditions, most classical models showed degraded performance. Nevertheless, the Rao et al. formulation (M-2) consistently emerged as the most reliable among the evaluated models across multiple locations. Although R^2 values during monsoon months remained low (typically 0.07–0.34), M-2 maintained comparatively lower RMSE values (approximately 3.5–5.0 MJ m⁻² day⁻¹) and near-zero MBE at several stations, demonstrating superior robustness against cloud-induced noise. This indicates that, even within the constraints of temperature-only inputs, certain formulations are better suited for monsoon applications.

Overall, the findings unequivocally emphasize that season-specific and region-specific model selection is essential for improving solar radiation estimation accuracy across India. Based on the comprehensive GPI-based evaluation, the following practical recommendations are proposed:

1. For inland regions and annual estimation: M-1 remains the most reliable and operationally simple choice.
2. For coastal and humid regions: M-5 and M-13 are preferable due to their improved response to humidity-influenced radiative conditions.
3. For monsoon season across India: M-2 provides the most robust and consistent performance among temperature-based models.
4. For plateau and central Indian regions: modified and non-linear temperature formulations (e.g., M-11 and M-13) offer enhanced sensitivity during winter and pre-monsoon periods.

This season-wise and region-wise framework offers a robust and nationally applicable basis for improving solar radiation estimation, particularly for Agrometeorological Advisory Services (AAS), crop simulation modelling, evapotranspiration estimation, and climate–energy planning. Importantly, the study confirms that several temperature-based empirical models can be reliably applied without local calibration, which is a substantial operational advantage in regions where ground-based radiation measurements are sparse or unavailable.

4.1 Limitations and Future Research Directions

The scientific rigor of this study necessitates acknowledgment of its inherent limitations, which also define clear pathways for future research. The primary limitations include the absence of local calibration and the use of Angström–Prescott model-derived radiation as the reference rather than direct pyranometric measurements. Consequently, the reported performance rankings reflect the intrinsic transferability and robustness of the evaluated models rather than their site-optimized accuracy.

4.2 Building upon this foundation, future research should focus on:

1. Local calibration and seasonal optimization of top-performing models to further enhance estimation accuracy.
2. Validation against high-quality pyranometric measurements to confirm model hierarchy and quantify absolute uncertainty.
3. Development of hybrid or machine-learning-based approaches that integrate the physical strengths of empirical models with data-driven pattern recognition.
4. Operational integration into decision-support systems, enabling automated selection of optimal models based on location and season.

In conclusion, despite specific methodological constraints, this study provides the first nationally consistent and operationally oriented framework for selecting temperature-based solar radiation models across India. By shifting the paradigm from generic model application to season- and region-specific selection, and by demonstrating the transferability of key models without local calibration, the study delivers an immediately actionable tool for enhancing agrometeorological advisories and climate services while establishing a solid foundation for future methodological advancements in solar radiation estimation research.

Declarations

Ethics Approval and Consent to Participate:

Not applicable. This research paper does not involve human participants or sensitive data requiring ethical approval.

Consent for Publication

Not applicable: This research paper does not contain any person's data in any form.

Use of AI

The authors used ChatGPT (OpenAI) and Deepseek to enhance the linguistic clarity and grammar of the manuscript. AI was also used to improve the visual resolution and formatting of figures. All substantive intellectual content, analysis, and conclusions remain the work of the authors.

Competing Interests:

The authors declare no competing interests.

Author Contribution

M.K. carried out data curation, methodological implementation, and prepared the first draft of the manuscript. A.K.M. was responsible for conceptualization, data acquisition, methodological design, and critical review and editing of the manuscript. K.G. contributed to project administration, validation of results, and manuscript review and editing. A.L. assisted with methodological development, manuscript review, and editing. All authors reviewed the manuscript.

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Data Availability:

The data used in this study are available from the India Meteorological Department (IMD), subject to institutional permissions.

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Figures

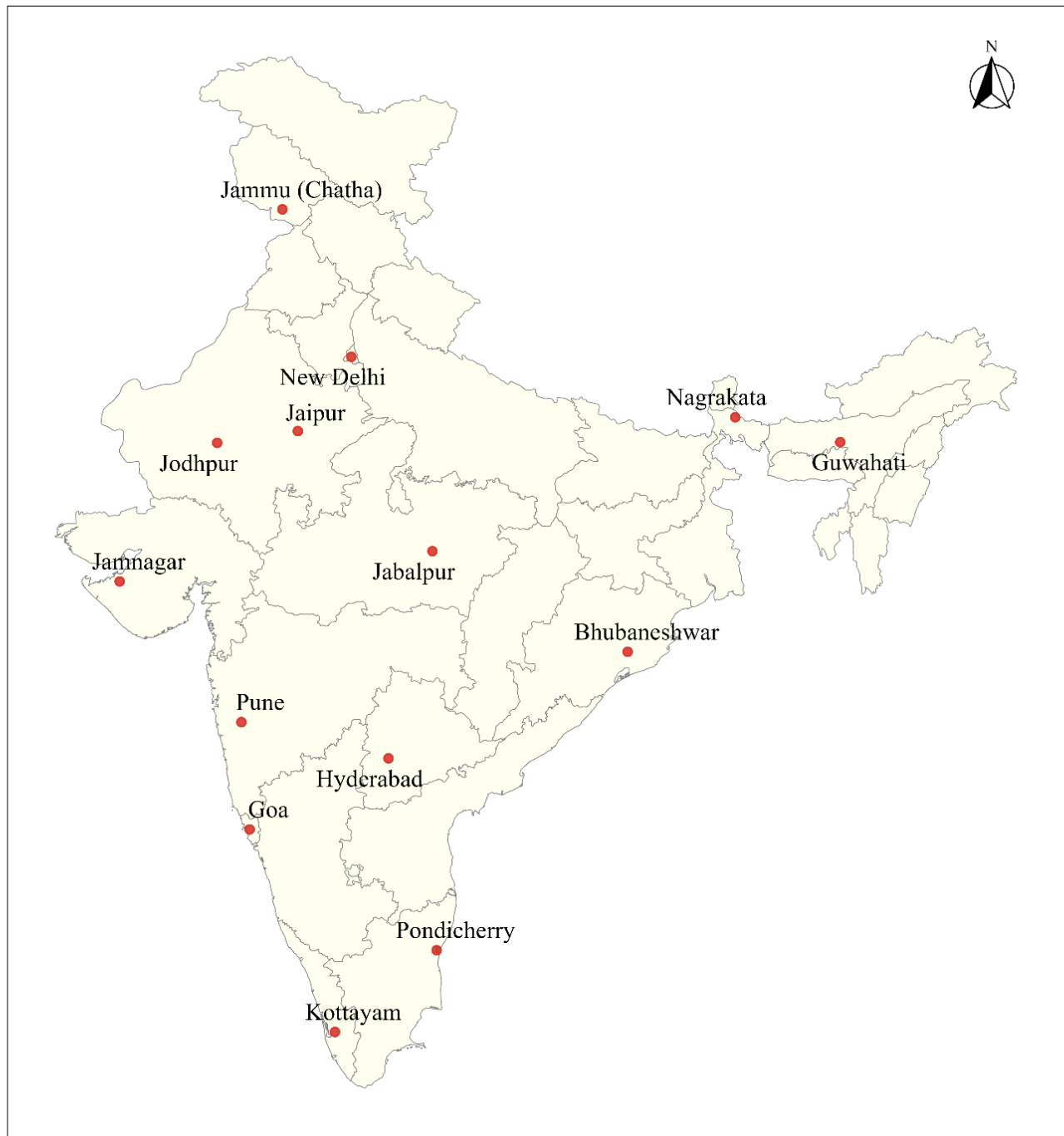


Figure 1

Spatial distribution of study stations across major climatic regions of India

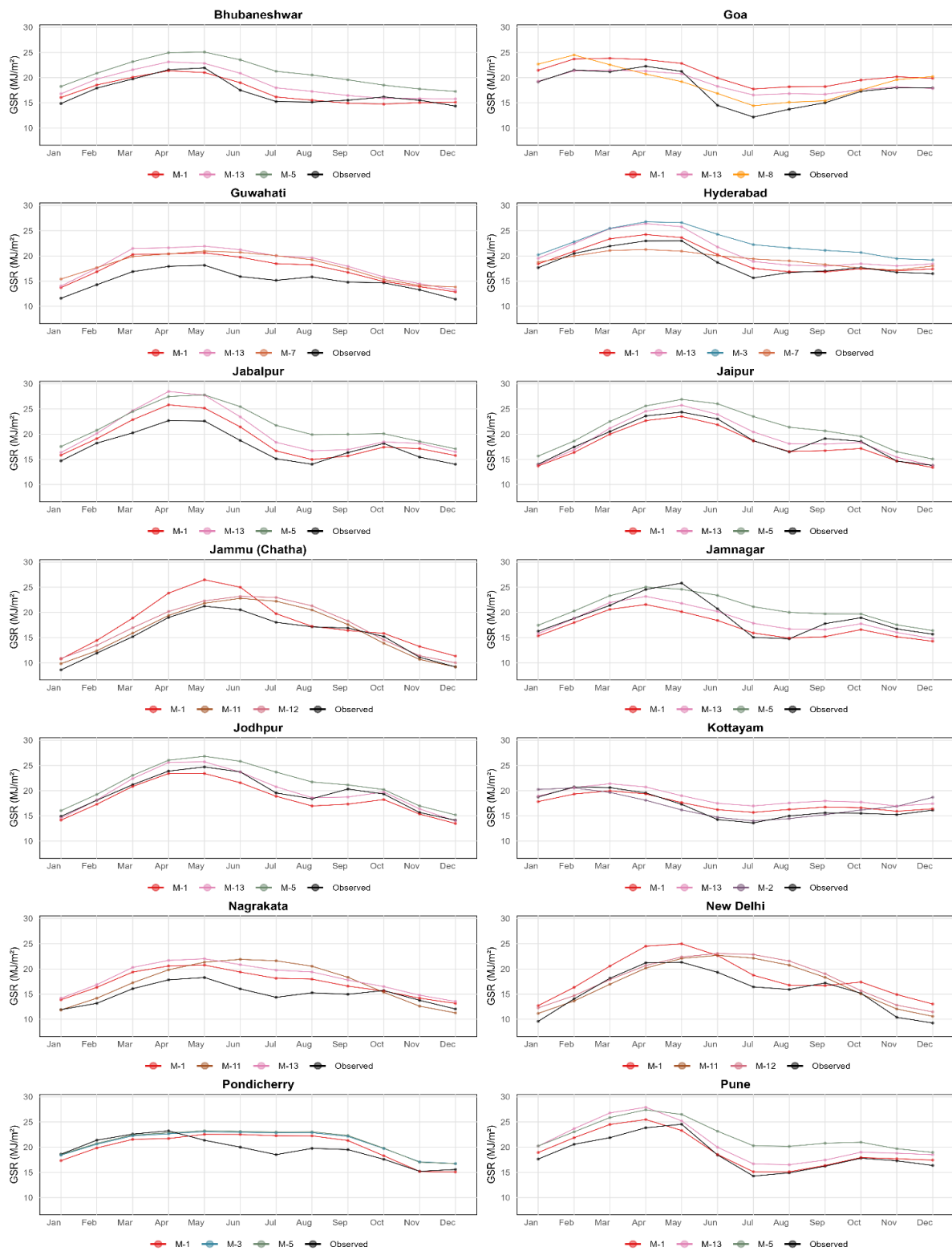


Figure 2

Monthly variation of solar radiation estimates from the top three performing temperature-based models compared with reference (Angstrom-Prescott) estimates for 14 representative locations across India